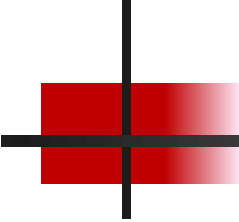
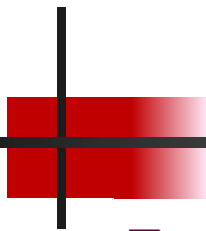


Denial of Service (DoS) & Distributed Denial of Service (DDoS) Attacks





Denial of Service (DoS) Attack

- ❑ A DoS attack is when a single machine or attacker floods a server or network with traffic
- ❑ Making it unavailable to legitimate users
- ❑ It overloads the system's resources (like CPU, memory, or bandwidth)
- ❑ Causing it to crash or slow down
- ❑ **Example:** A hacker sends an enormous number of requests to a website, exhausting its resources so that normal users can't access it

Distributed Denial of Service (DDoS) Attacks

- ❑ A **DDoS attack** involves overwhelming a system or network with traffic from multiple sources to make it unavailable to legitimate users.
- ❑ **Mechanism:**
 - Uses a **botnet** of compromised systems (zombies)
 - Launches massive traffic toward a target (server, website, etc.)
 - Can cause **downtime**, financial loss, and **service unavailability**

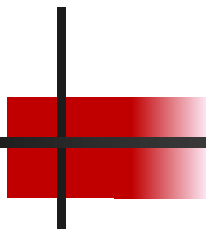
Distributed Denial of Service (DDoS) Attacks

□ Types of DDoS Attacks:

- **Volume-based** – Consumes bandwidth (e.g., UDP floods)
- **Protocol attacks** – Exploit server resources (e.g., SYN floods)
- **Application-layer** – Target specific apps (e.g., HTTP GET/POST flood)

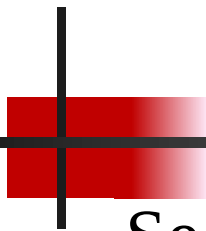
□ Prevention Techniques:

- Intrusion detection systems (IDS)
- Firewalls and rate limiting
- Traffic filtering and scrubbing services (analyze, filter, and clean traffic)
- Load balancing and redundancy



DoS vs. DDoS

Feature	DoS	DDoS
Origin	Single attacker	Multiple attackers (botnet)
Complexity	Easier to detect & block	Harder to trace and mitigate
Power	Limited impact	Massive, large-scale disruption
Cost	Less costly to launch	Requires control of many devices



Exercise

SecureFinance recently suffered a major operational collapse. First, their client portal was overwhelmed by a DDoS attack, where a botnet of "zombies" flooded the server with traffic. Simultaneously, an employee accidentally triggered a virus by executing a malicious file disguised as a report, which then began encrypting sensitive data.

Question: Identify the three specific types of malware or attacks mentioned above. Compare the DDoS attack to a standard DoS attack in terms of origin and complexity. Also, explain why the file-encrypting software is classified as an Active Threat rather than a Passive Threat. List three specific techniques used to prevent or mitigate the impact of a DDoS attack. Explain the difference between Volume-based and Application-layer DDoS attacks.